

Is it the grain size or the characteristic pore size that controls the induced polarization relaxation time of clean sands and sandstones?

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[1] There is a wide range of evidence to suggest that permeability can be constrained through of induced polarization measurements. For clean sands and sandstones, current mechanistic models of induced polarization predict a relationship between the low-frequency time constant inferred from induced polarization measurements and the grain diameter. A number of observations do, however, disagree with this and indicate that the observed relaxation behavior is rather governed by the so-called dynamic pore radius Λ . To test this hypothesis, we have developed a set of new scaling relationships, which allow the relaxation time to be computed from the pore size and the permeability to be computed from both the Cole-Cole time constant and the formation factor. Moreover, these new scaling relationships can be also used to predict the dependence of the Cole-Cole time constant as a function of the water saturation under unsaturated conditions. Comparative tests of the proposed new relationships with regard to various published experimental results for saturated clean sands and sandstones as well as for partially saturated clean sandstones, do indeed confirm that the dynamic pore radius Λ is a much more reliable indicator of the observed relaxation behavior than grain-size-based models.

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1. Introduction

[2] Frequency domain induced polarization is a low-frequency geoelectrical method based on the use of an alternating harmonic current and the measurement of the amplitude and phase lag of the associated electrical field. These measurements can then be interpreted in terms of a frequency-dependent complex electrical conductivity with an in-phase component controlled by electro-migration of the charge carriers through the pore network as well as along the grain/water interfaces of the probed porous material and an out-of-phase or quadrature component related to polarization phenomena associated with the temporary reversible storage of electrical charges [e.g., *Weller and Börner*, 1996; *Binley et al.*, 2005]. The low-frequency complex conductivity of clean sands and sandstones, inferred from frequency domain induced polarization measurements typically exhibits a characteristic dispersive behavior, which

can be described by a Cole-Cole-type distribution model [e.g., *Cole and Cole*, 1941].

[3] A long-standing, fundamental, and as of yet unresolved question is which textural parameters control the observed relaxation times in porous media [*Kruschwitz et al.*, 2010]. In colloids and granular media, models for the polarization of the electrical double layer around the grains imply that the controlling parameter is the grain size or the grain size distribution [e.g., *Schwarz*, 1962; *Grosse*, 2009; *Revil and Florsch*, 2010]. Conversely, the protagonists of the membrane polarization model argue that the pore length is the characteristic scale governing the diffusion-based relaxation mechanisms [e.g., *Titov et al.*, 2002, 2004]. For simple granular porous media, such as clean sands and sandstones, the pore length is roughly equal to the grain diameter [*Bernabé et al.*, 2011] and hence both types of models essentially predict a scaling of the relaxation time with regard to the grain diameter. A number of observations do, however, disagree with these models and point toward a control of the relaxation times by the so-called characteristic pore size of the porous material that can be assessed by Hg porosimetry [e.g., *Scott and Barker*, 2003; *Kruschwitz et al.*, 2010] or by other pore size parameters like some effective pore throat radius as proposed below. This observation is hard to reconcile with existing models as it would imply a polarization direction perpendicular to the direction of the applied electrical field in the pore space. If the throats are short, as proposed in the short pore model of *Titov et al.* [2002], the pore throat radius may scale with the pore length. This idea needs to be further explored through pore scale modeling and observations made on thin sections.

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[4] In the present work, we explore the importance of the mean grain diameter and the characteristic size of the pore throat with regard to the relaxation time of clean sands and sandstones (the modeling of the induced polarization response of shaly sands and sandstones has been investigated recently by *Revil* [2012]). To this end, we derive new relationships to predict the relaxation time. We then test these new relationships on previously published experimental data for saturated clean quartz sands as well as clean sandstones and compare the corresponding predictions with those by a grain-size-based model.

2. Background

2.1. Saturated Case

[5] We define the following canonical boundary value problem for the normalized potential Γ for a cylindrical representative elementary volume of porous material of length L

$$\nabla^2 \Gamma = 0 \text{ in } V_p, \quad (1)$$

$$\hat{n} \cdot \nabla \Gamma = 0 \text{ on } S, \quad (2)$$

$$\Gamma = \begin{cases} L & \text{at } z = L \\ 0 & \text{at } z = 0 \end{cases} \text{ on } S, \quad (3)$$

where $\hat{n}$ is the unit vector normal to the pore water/mineral interface S and V_p denotes the pore volume. These boundary conditions are representative for the injection of an electrical current into a rock sample in the absence of an electrical double layer at the pore/water interface [e.g., *Johnson et al.*, 1986] and therefore in absence of polarization. The dynamic pore radius Λ and the formation factor F are defined as [*Johnson et al.*, 1986; *Avellaneda and Torquato*, 1991]

$$\frac{2}{\Lambda} = \frac{\int_S |\nabla \Gamma|^2 dS}{\int_{V_p} |\nabla \Gamma|^2 dV_p}, \quad (4)$$

$$\frac{1}{F} = \frac{1}{V} \int_{V_p} |\nabla \Gamma|^2 dV_p, \quad (5)$$

where V is the total volume of the considered representative elementary volume. From equation (2), Λ is a weighted version of the hydraulic pore radius V_p/S , the weight being the norm of the normalized electrical field existing in the pore space in the absence of an electrical double layer. Because this electrical field is strong in the throats and negligible in the dead-ends, Λ represents a dynamic pore throat radius for the porous material and hence can be regarded as the characteristic pore size. Equation (5) indicates that $1/F$ can be regarded as a dynamic porosity [*Revil and Cathles*, 1999]. Note that all the parameters Λ , F , V_p/S , and ϕ are measurable properties [e.g., *Johnson et al.*, 1986; *Revil and Cathles*, 1999]. The permeability k (in units of m^2) is then related to the formation factor F and the dynamic pore radius Λ as [*Johnson et al.*, 1986; *Avellaneda and Torquato*, 1991]

$$k = \frac{\Lambda^2}{8F}, \quad (6)$$

Revil and Cathles [1999] showed that equation (6) offers a better predictive equation than the Kozeny-Carman model based on the specific surface area and used in many studies to predict the permeability from geophysical measurements [e.g., *Börner et al.*, 1996]. The formation factor F is also related to the connected porosity ϕ by $F = \phi^{-m}$ with m denoting the cementation exponent [*Archie*, 1942]. *Revil and Cathles* [1999] developed a relationship between the formation factor F , the dynamic pore radius Λ , and the grain diameter d for the case of an ideal granular material, in which all grains are spherical and have the same diameter [e.g., *Bolève et al.*, 2007]

$$\Lambda = \frac{d}{2m(F-1)}, \quad (7)$$

where m is again the cementation exponent. Equation (7) is valid for clean sands and sandstones with a unimodal grain size distribution. Note that this expression can be easily generalized in the presence of a range of grain sizes [*Revil and Florsch*, 2010] using the expectation of the probability density for the inverse of the grain size.

[6] Once corrected by a geometrical factor accounting for the experimental setup, the resistance and the phase lag observed in the course of frequency domain induced polarization measurements can then be expressed in terms of a complex electrical conductivity of the form

$$\sigma^*(\omega) = |\sigma(\omega)| \exp(i\varphi(\omega)), \quad (8)$$

where ω is the angular frequency and $i = \sqrt{-1}$ the imaginary number. The model developed by *Revil and Florsch* [2010] uses a Cole-Cole probability density function for the distribution of the relaxation times and can be written as [*Cole and Cole*, 1941]

$$\sigma^*(\omega) = \sigma_\infty \left[1 - \frac{M}{1 + (i\omega\tau_{CC})^c} \right] + i\omega\varepsilon_\infty, \quad (9)$$

where τ_{CC} is the Cole-Cole time constant, and c is the dimensionless Cole-Cole exponent ranging between 0.5 and 0, with $c = 0.5$ corresponding to the idealized case of all grains having the same diameter, ε_∞ is the high-frequency dielectric constant, and $M = (\sigma_\infty - \sigma_0)/\sigma_\infty$ is the dimensionless chargeability with σ_0 and σ_∞ denoting the DC ($\omega = 0$) and high-frequency ($\omega \gg 1/\tau_{CC}$) electrical conductivities, respectively. The low- and high-frequency components entering into equation (9) are given by [*Revil and Florsch*, 2010]:

$$\sigma_0 = \frac{1}{F} \left[\sigma_w + \frac{2}{\Lambda} \Sigma_d \right], \quad (10)$$

$$\sigma_\infty = \frac{1}{F} \left[\sigma_w + \frac{2}{\Lambda} (\Sigma_d + \Sigma_S) \right], \quad (11)$$

$$\varepsilon_\infty = \frac{1}{F} [\varepsilon_w + (F-1)\varepsilon_S], \quad (12)$$

where Σ_d and Σ_S denote the specific surface conductivities of the diffuse and Stern layers, respectively, σ_w and ε_w

denote the electrical conductivity and dielectric constant of the pore water, and ε_s denotes the dielectric constant of the solid phase. From equations (10) and (11), the chargeability is approximately given by $M \approx (2/\Lambda) (\Sigma_s / \sigma_w)$. Note that Σ_s depends on salinity as modeled by *Revil and Florsch* [2010] and *Revil and Skold* [2011].

[7] The angular frequency of the phase peak is given by

$$\omega_{\text{peak}} = \left(\frac{\sigma_\infty}{\sigma_0} \right)^{1/2c} \frac{1}{\tau_{CC}} = \left(\frac{1}{1-M} \right)^{1/2c} \frac{1}{\tau_{CC}}. \quad (13)$$

If the chargeability M is very small, which is indeed the case for the clean saturated sands investigated in the following (Table 1), we have $\omega_{\text{peak}} \approx 1/\tau_{CC}$. Note that there are alternative approaches in fitting polarization spectra to avoid the use of a specific parametric function. For example, *Nordsiek and Weller* [2008] and *Tong et al.* [2006] used Debye decomposition approaches in the frequency- and time-domains, respectively.

[8] In the induced polarization model developed by *Revil and Florsch* [2010], the Cole-Cole time constant can be approximately related to the grain diameter d by

$$\tau_{CC} \approx \frac{d^2}{8D_i}. \quad (14)$$

The coefficient D_i represents the diffusion coefficient of the counterions, generally cations. This coefficient is temperature-dependent with an increase of $\sim 2\%$ per degree Kelvin. In the work of *Koch et al.* [2011], the pore fluid was a NaCl electrolyte with $D_+ = 1.32 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ at 25°C and the mobility of the cations given by $\beta_+ = 5.19 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$.

[9] From the point of view of dimensional analysis, equation (14) corresponds to a simple scaling law between the diffusion coefficient D_i in units of $\text{m}^2 \text{ s}^{-1}$, a relaxation time τ_{CC} in units s, and the squared pore diameter d^2 in

units of m^2 . Another acceptable scaling law from the point of view of dimensional analysis would be

$$\tau_{CC} = \frac{\Lambda^2}{2D_i}, \quad (15)$$

(the factor of 2 in equation (15) is obtained by replacing the grain diameter by its relationship to the grain radius and stating that the polarization length scale is not the grain radius but the pore throat size Λ). Equation (15) implies that the dynamic pore radius Λ corresponds to the fundamental scaling parameter for the relaxation time. In other words, Λ is assumed to be the characteristic polarization length scale of the porous material. At this point, equation (15) is still in need of a mechanistic model to justify its existence. That said polarization of the electrical double layer is able to polarize the adjacent pore water in the standard model for colloidal suspensions [*Grosse*, 2009]. Equation (15) is also consistent with the observations made by *Scott and Barker* [2003] and *Krushwitz et al.*, [2010]. They observed that for granular material with large pore sizes ($>10 \mu\text{m}$), a positive correlation exists between the time constant and the pore throat size. This observation is also consistent with the data set of *Binley et al.* [2005] as discussed below. Equations (14) and (15) both predict that the time constant is nearly independent on the salinity as shown by *Revil and Skold* [2011] and *Weller et al.* [2011] (a small salinity dependence in equations (14) and (15) can arise because the diffusion coefficient may be slightly dependent on salinity).

[10] In the following, we consider that equation (15) is our starting hypothesis and we are going to build additional relationships based on this concept. Replacing Λ in equation (15) with its expression in equation (7), we obtain

$$\tau_{CC} = \frac{d^2}{4m^2(F-1)^2 D_i}. \quad (16)$$

Table 1. Grain Size Characteristics of Loose and Compacted Quartz Sands as Inferred from Laser Diffraction Measurements^a

Sample	d_{50} (mm)	k (m^2)	F	ϕ	m	M^b	SD (log)	τ_{CC} (s)	SD (log)	Λ (μm) ^c
F36	0.18	1.76E-11	3.77	0.44	1.59	0.0119	0.158	0.44	0.065	20.4 ± 1.6
F32	0.27	5.31E-11	3.55	0.44	1.54	0.0022	0.035	0.51	0.030	34.4 ± 2.8
WQ1	0.66	1.29E-10	3.25	0.47	1.56	0.0232	0.060	2.13	0.054	94.0 ± 7.8
SP1	0.18	2.08E-11	3.14	0.46	1.48	0.0048	0.051	0.30	0.065	28.4 ± 2.3
SP2	0.24	3.30E-11	3.40	0.44	1.49	0.0066	0.046	0.30	0.094	33.6 ± 2.7
SP3	0.32	6.75E-11	3.26	0.49	1.65	0.0051	0.103	5.13	0.042	42.9 ± 3.4
SP4	0.50	1.71E-10	3.12	0.49	1.58	0.0063	0.048	0.84	0.044	74.6 ± 6.0
SP5	0.68	2.80E-10	3.10	0.48	1.54	0.0121	0.077	4.68	0.163	105 ± 8.4
SP6	0.87	3.94E-10	3.34	0.49	1.70	0.0113	0.096	12.4	0.173	109 ± 8.7
F36-C	0.18	1.11E-11	4.12	0.38	1.48	0.0232	0.363	0.23	0.507	19 ± 1.5
F32-C	0.27	2.40E-11	3.75	0.39	1.40	0.0038	0.057	0.14	0.167	35.1 ± 2.8
WQ1-C	0.66	7.50E-11	3.97	0.42	1.59	0.0082	0.056	1.86	0.062	69.9 ± 5.6
SP1-C	0.18	1.17E-11	3.23	0.41	1.30	0.0068	0.059	0.14	0.005	31.0 ± 2.5
SP2-C	0.24	1.98E-11	3.55	0.39	1.35	0.0050	0.055	0.40	0.058	34.9 ± 2.8
SP3-C	0.32	3.81E-11	3.64	0.42	1.49	0.0073	0.042	5.49	0.044	40.7 ± 3.3
SP4-C	0.50	1.05E-10	3.52	0.44	1.54	0.0087	0.059	0.80	0.091	64.4 ± 5.2
SP5-C	0.68	1.96E-10	3.36	0.43	1.44	0.0122	0.072	3.65	0.160	100 ± 8.0
SP6-C	0.87	2.56E-10	3.63	0.43	1.53	0.0080	0.115	3.42	0.409	108 ± 8.6

^aSee *Koch et al.* [2011]. The grain size d_{50} denotes the median of the particle size distribution and is the length scale used for the grain diameter d . The electrical measurements were taken at temperature of 21.6°C . Samples denoted as “-C” represent compacted sands. The cementation exponent m is determined as $m = -\ln F / \ln \phi$ and the length scale Λ from equation (7) using the values of d_{50} , m , and F . The uncertainty of F is of the order of 3%, the uncertainty of ϕ is of the order of 2%, and the uncertainty of m is of the order of 5%. The Cole-Cole parameters have all been obtained by fitting equation (9) to the data.

^bA Turn brine conductivity (NaCl) $0.004\text{--}0.006 \text{ S m}^{-1}$.

^cDetermined from equation (7). Relative error determined from $\frac{\Delta \Lambda}{\Lambda} = \frac{\Delta d_{50}}{d_{50}} + \frac{\Delta m}{m} + \frac{\Delta F}{F}$.

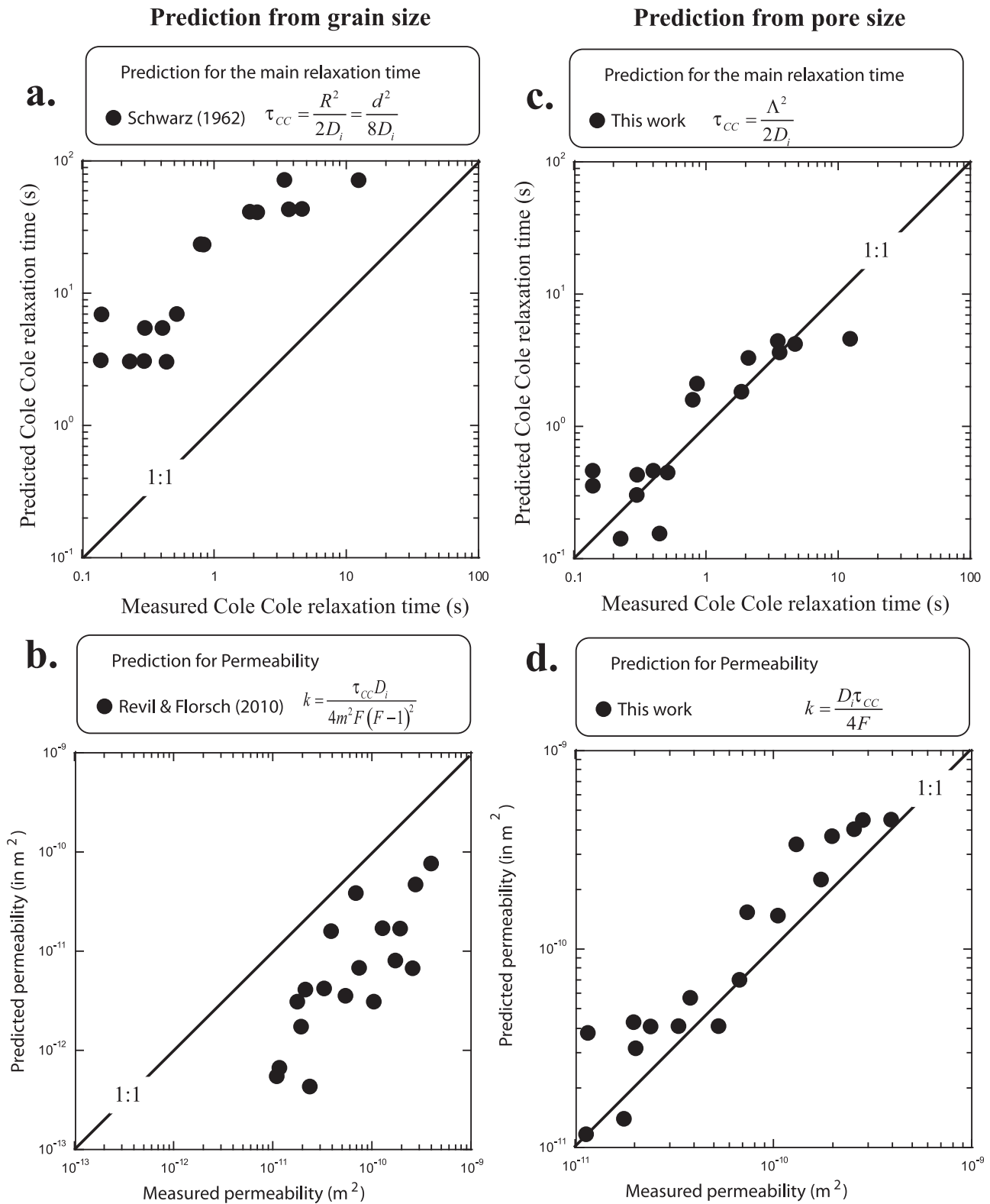


Figure 1. Predicted versus measured parameters from the grain size and the pore size for the clean saturated sand samples list in Table 1 [Koch *et al.* 2011]. Test of the grain size model for the Cole-Cole time constant (a) (powerlaw fit: $R^2 = 0.70$) and for the permeability (b) (powerlaw fit: $R^2 = 0.61$). Test of the pore size model for the Cole-Cole time constant (c) (powerlaw fit: $R^2 = 0.69$) and for the permeability (d) (powerlaw fit: $R^2 = 0.88$).

Equation (16) predicts a very specific dependence of the Cole-Cole relaxation time τ_{CC} with the formation factor F and therefore with the connected porosity ϕ and, as shown later, with the water saturation. This also means that the Cole-Cole relaxation time τ_{CC} increases with compaction. This is consistent with the measurements reported in Table 1 and the results of previous work [Klitzsch, 2004; Halisch and Weller, 2006]. We can now follow previous attempts and use the time constant to determine the permeability [e.g., Titov et al., 2010; Revil and Florsch, 2010]. Substituting equation (15) into equation (6) yields the following expressions for the permeability

$$k = \frac{D_i \tau_{CC}}{4F}. \quad (17)$$

The limitations of equation (17) are the following: (1) we are dealing only with clean sands and clean sandstones (for clayey sands see Revil [2012]), (2) the pore size distribution needs to be unimodal and reasonably symmetric for the Cole-Cole model to apply, and (3) the pore size is $>10\text{--}15\text{ }\mu\text{m}$ or so. In the following, we shall test the newly developed relationships based on the characteristic pore size with regard to two set of laboratory measurements and compare the corresponding predictions with those of the grain-size-based model of Revil and Florsch [2010].

2.2. Unsaturated Case

[11] By analogy to Archie's first and second laws [Archie, 1942] and in agreement with the Vinegar and Waxman [1984] dependence of surface conductivity with saturation, we can perform the following change of variables $(1/F) \Rightarrow (1/F)s_w^n$ and $\Lambda \Rightarrow \Lambda s_w^n$ (with s_w denoting the water saturation and n the saturation exponent) to extend our model to the general case of partial saturation. This yields [e.g., Linde et al., 2006]

$$\sigma_0 = \frac{1}{F} \left[s_w^n \sigma_w + \frac{2}{\Lambda} \Sigma_d \right], \quad (18)$$

$$\sigma_\infty = \frac{1}{F} \left[s_w^n \sigma_w + \frac{2}{\Lambda} (\Sigma_d + \Sigma_s) \right], \quad (19)$$

$$\varepsilon_\infty = \frac{1}{F} [s_w^n \varepsilon_w + (1 - s_w^n) \varepsilon_a + (F - 1) \varepsilon_s], \quad (20)$$

where ε_a denotes the dielectric constant of air. The chargeability should be given by $M(s_w) \approx M(s_w = 1)s_w^{-n}$ and therefore should increase with decreasing saturation. The same change of variables yields the following scaling relationship between the Cole-Cole time constant and the saturation: $\tau_{CC}(s_w) \approx \tau_{CC}(s_w = 1)s_w^{2n}$, where $\tau_{CC}(s_w = 1)$ represents the value of the Cole-Cole time constant at full water saturation. The exponent $2n$ is justified because the Cole-Cole time constant is roughly inversely proportional to square of the formation factor when $F \gg 1$ and, more directly, by the used change of variables $\Lambda \Rightarrow \Lambda s_w^n$ in equation (15).

3. Tests on Laboratory Measurements

3.1. Frequency-Domain Induced Polarization Measurements on Clean Saturated Sands

[12] The frequency domain induced polarization measurements considered here were acquired in a frequency

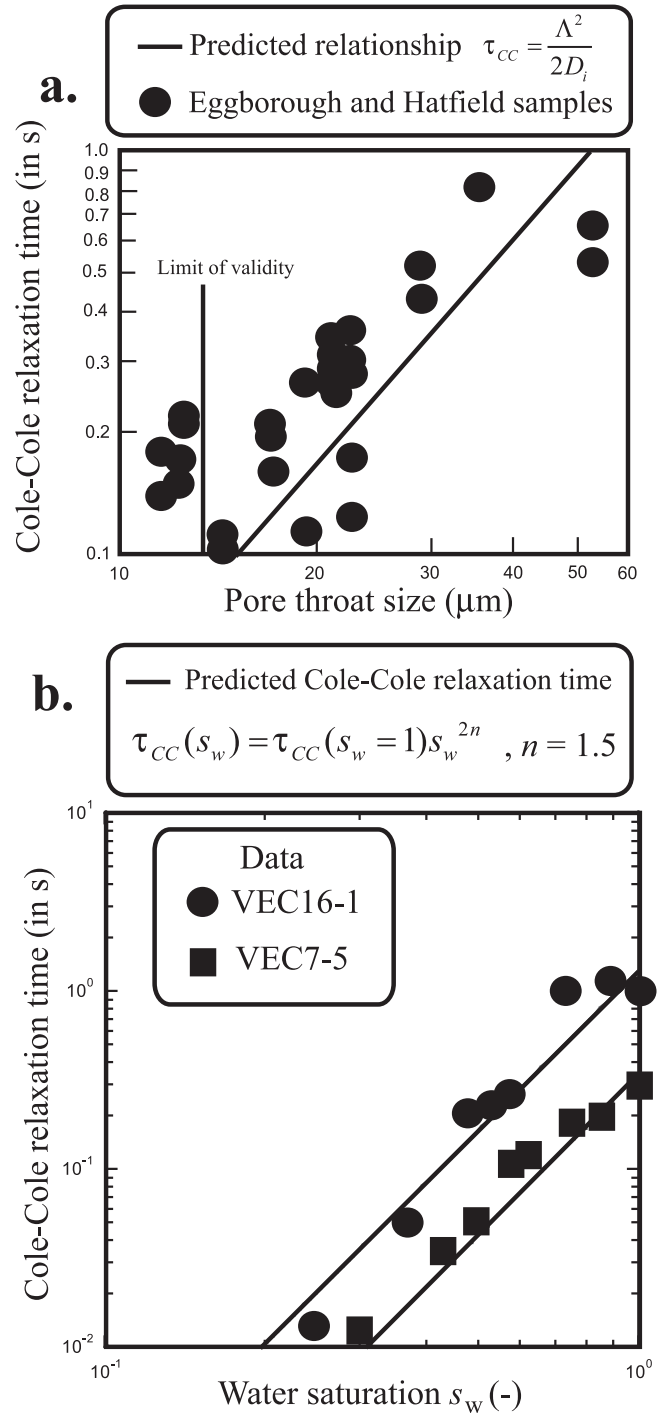


Figure 2. Influence of pore size and saturation on the Cole-Cole time constant. (a) Comparison between the predicted and measured Cole-Cole time constant for clean parts of the Sherwood sandstone (see Binley et al. [2005], their Figure 8, $R^2 = 0.66$ excluding the low pore sizes below $15\text{ }\mu\text{m}$). (b) Comparison between the prediction of the model for the dependence of the Cole-Cole time constant with the saturation and the experimental data of Binley et al. [2005] (see their Figure 13). Sample VEC16-1: $R^2 = 0.74$; Sample VEC7-5: $R^2 = 0.90$.

range from 1 mHz to 45 kHz [Koch *et al.*, 2011] using a highly sensitive impedance spectrometer with an accuracy of 0.1 mrad for the phase for frequencies between 1 mHz and 100 Hz. The samples were saturated with a NaCl electrolyte with an electrical conductivity σ_w ranging 0.004 and 0.006 S m⁻¹. The properties of the considered samples are listed in Table 1. To fit the Cole-Cole model to the observed data, we used a Bayesian Markov-chain Monte Carlo (McMC) sampler [Koch *et al.*, 2011]. All the samples from Koch *et al.* [2011] were considered with the exception of sample SP3, which contains plate-like particles of mica with a length of approximately one millimeter.

[13] A comparison between the prediction of the model of Revil and Florsch [2010] and the measured Cole-Cole relaxation times is shown in Figure 1a, whereas the corresponding comparison for equation (17) is shown in Figure 1c. There is a good agreement between the experimental data of Table 1 and the prediction of equation (17) and but not with regard to the model of Revil and Florsch [2010]. We then proceed compare the measured permeability (Table 1) with predicted permeability using the model of Revil and Florsch [2010] (Figure 1b) and equation (17) (Figure 1d). Again, the model based on the characteristic pore size (equation (17)) developed in this study provides a more favorable prediction than the grain-size-based model of Revil and Florsch [2010].

3.2. Partially Saturated Sandstones

[14] The second test of the newly proposed relationships concerns the dependence of the Cole-Cole relaxation time on the saturation. If the proposed scaling relationship between the Cole-Cole relaxation time and the formation factor is correct, we can easily predict the dependence of the Cole-Cole relaxation time on the saturation. We selected the data set of Binley *et al.* [2005] because it corresponds to

a clean sandstone. The Cole-Cole time constant for this data set is pretty well predicted by our model (Figure 2a) except for very small pore sizes (<15 μ m).

[15] A comparison between $\tau_{CC}(s_w) \approx \tau_{CC}(s_w = 1)s_w^{2n}$ and the experimental data of Binley *et al.* [2005] is shown in Figure 2b using a saturation exponent of $n = 1.5$, which is realistic for sands and sandstones with low clay content (typically in the range 1.2–1.6) [Vinegar and Waxman, 1984]. There is a good agreement between the observed and the predicted data.

[16] A comparison between $M(s_w) \approx M(s_w = 1)s_w^{-n}$ and the experimental data of Titov *et al.* [2004] is shown in Figure 3a. For this database, the saturation exponent n is determined from the saturation dependence of the resistivity data (Figure 3b) and is equal to $n = 1.3$. There is a fairly good agreement between the model and the experimental data down to a water saturation of 0.08. Below this critical water saturation, our model is not consistent with the data anymore and it is conceivable that the charge of the air/water interface is playing a role in the polarization mechanism.

4. Conclusions

[17] Extending existing models, we have developed a new set of scaling relationships, which indicate that the induced polarization relaxation characteristics of clean sands and sandstones may be governed by the dynamic pore radius rather than by grain diameter. These relationships were then tested on three different datasets from frequency domain polarization laboratory experiments and the results were compared to those obtained with a grain-size-based model. Our results indicate that the Cole-Cole time constant does indeed seem to be controlled by the dynamic pore radius rather than by the grain diameter, which in turn

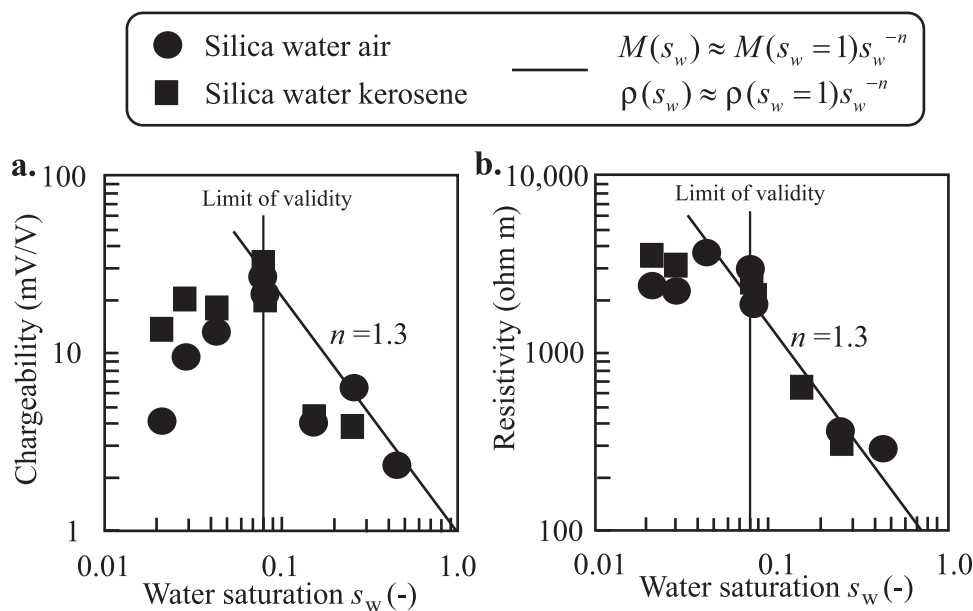


Figure 3. Observed [Titov *et al.*, 2004] versus predicted values of the saturation dependence of the chargeability and resistivity for silica sands with a saturation exponent n of 1.3. (a) Predicted chargeability dependence on saturation using the second Archie's law. (b) Predicted resistivity dependence on saturation using the second Archie's law.

supports the simple but novel scaling relationship between the permeability, the formation factor, and the time constant proposed in this study. This in turn does, however, also point to the need of finding an adequate mechanistic polarization model to explain these observations and to clarify why the dynamic pore radius Λ appears as a polarization length scale.

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